Comprehensive Peritoneal Fibrosis Discovery & Validation Workflow

GSE125498 Discovery Cohort (N=33) | Naba Human Matrisome (1,027 Genes) | GSE62928 Validation (N=8)

Phase 1: Discovery & Preprocessing

- Cohort: GSE125498
- N=33 (13 LPD vs 20 SPD)
- Illumina HumanHT-12
- 11,741 Collapsed Genes
- Max-mean probe collapse

Phase 2: Differential Expression (DEGs)

- $|\log_2 FC| \geq 0.585$
- Nominal $P < 0.05$
- NO FDR correction
- 813 Total DEGs
- 496 Up | 317 Down

Phase 3: Naba ECM Intersection

- Naba Matrisome (1,027)
- 428 Measured Genes
- 44 Matrisomal DEGs
- 31 Pro-fibrotic Up
- Hypergeom P = 0.0024

Phase 4: WGCNA Network Analysis

- 5,001 Variable Genes
- 14 Modules Detected
- Top Module: MEmagenta (r = 0.48, P = 0.0047)
- Hub cluster in MEblack

Phase 5: Ensemble ML Prioritization

- LASSO (10 selected)
- SVM-RFE (9 selected)
- Random Forest (9 sel)
- XGBoost (9 selected)
- 11 Consensus Hubs

11 Consensus Hub Biomarkers

Top Regulators:

- FLT3LG (logFC = 1.44, P = 2.4e-5) [4/4 votes]
- TNFSF15 (logFC = 0.97, P = 2.7e-5) [4/4 votes]
- ADAM19 (logFC = 1.21, P = 1.3e-4) [3/4 votes]
- SERPINA10 (logFC = 1.17, P = 6.5e-4) [3/4 votes]
- LTB, CLEC4F, SEMA3E, SEMA4F, EBI3, LTA, CXCL14

→ All 11 hubs up-regulated in LPD fibrosis

External Validation (GSE62928)

Independent Human Cohort:

- 4 Encapsulating Peritoneal Sclerosis (EPS)
- 4 Normal Peritoneal Controls
- Apparent ROC AUC = 1.000
- Leave-One-Out CV AUC = 1.000
- 100% Diagnostic discrimination

→ Robust cross-cohort transferability

Clinical Translation & Networks

Clinical Nomogram & Scoring:

- Apparent C-index = 0.981
- 10-Fold CV C-index = 0.873
- High net clinical benefit on DCA

Functional Interplay:

- LTB-LTA-TNFSF15 cytokine triad
- Myofibroblast score elevation (P = 0.36)